Geographic Variation in the Black-White Life Expectancy Gap in Wisconsin, 2005-2017

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Abstract

Racial disparities in life expectancy persist across the United States, but state-level averages can mask important geographic heterogeneity. This paper examines the Black-White life expectancy gap in Wisconsin from 2005 to 2017, disaggregating the state into Milwaukee County, Dane County (Madison), and the rest of the state. Because Milwaukee is home to about two-thirds of Wisconsin's Black population and exhibits very poor Black mortality outcomes, statewide statistics are heavily influenced by conditions in a single county. I construct abridged life tables by race, sex, geography, and period, validate statewide estimates against prior work, and apply a formal decomposition that expresses the statewide gap as the sum of geography-specific contributions (population share times local gap), along with a composition term and residual. For males in 2015--2017, Milwaukee accounts for about 70 percent of the 7.9-year statewide gap, and the geographic concentration of Black Wisconsinites in Milwaukee adds roughly 2.6 years to the gap relative to a scenario in which Black and White residents share the same distribution across regions. Counterfactual exercises show that if Black Milwaukeeans had the same age-specific mortality as Black residents in the rest of Wisconsin, the statewide male gap would fall from 7.9 to about 4.2 years. In Dane County, a large male gap (8.5 years) reflects both exceptionally high White life expectancy and evidence of worsening Black male mortality, although the latter is estimated with substantial uncertainty because of the small Black population. Bootstrap confidence intervals quantify this uncertainty and highlight that the strongest and most precisely estimated contribution to the statewide gap arises from Milwaukee's combination of concentrated Black population and severe local mortality disadvantage.

Keywords: life expectancy, racial disparities, Wisconsin, geographic variation, counterfactual analysis

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1. Introduction

The Black-White gap in life expectancy is one of the most persistent health disparities in the United States. Although this gap has narrowed nationally over recent decades, substantial variation remains across states and localities (Harper et al. 2014; Riddell et al. 2018). State-level analyses, while informative, can obscure important within-state heterogeneity driven by residential segregation, differential access to health care, economic inequality, and other social determinants of health.

Wisconsin is a particularly instructive case. The state consistently ranks among the worst for Black-White disparities across a range of social and economic indicators. The Milwaukee metropolitan area has been repeatedly identified as one of the most racially segregated in the nation. Roberts et al. (2019) documented a persistent statewide Black-White life expectancy gap in Wisconsin that showed little improvement between 1999 and 2016, in contrast to national trends of convergence. Critically, however, their analysis was conducted at the state level.

A state-level analysis of Wisconsin is misleading for a specific, quantifiable reason: Milwaukee County alone contains about two-thirds of the state's Black population (67 percent; see Appendix C). Wisconsin is not unique in this respect: in many states, a single county holds the largest share of the state's Black population (see Appendix C). If mortality conditions for Black residents of Milwaukee are substantially worse than elsewhere in the state -- as one might expect given the city's concentrated poverty and extreme segregation -- then the statewide Black life expectancy is overwhelmingly a Milwaukee statistic. The "Wisconsin gap" is, to a considerable extent, the Milwaukee gap. Conversely, Dane County (Madison) has experienced rapid growth in its Black population and is often seen as a high-opportunity community, yet casual observation of aggregate statistics may overstate racial equity there.

This paper makes three contributions. First, I disaggregate Wisconsin's Black-White life expectancy gap into three regions -- Milwaukee County, Dane County, and the rest of the state -- revealing striking geographic variation. Second, I show that the statewide gap is dominated by the combination of very poor Black mortality conditions in Milwaukee and the fact that about two-thirds of Wisconsin's Black population lives there, yielding a substantial composition penalty. Third, I use counterfactual mortality-replacement exercises to quantify how much of the statewide gap would remain if Milwaukee's Black mortality matched that of Black residents in the rest of the state or if Dane County's White mortality matched that of Whites elsewhere. The analytic approach is validated against the published results of Roberts et al. (2019); this replication is reported in the appendix.

2. Background

A growing literature documents within-state and within-city variation in racial life expectancy gaps. Murray et al. (2006) showed that life expectancy varied enormously across population subgroups defined by geography and race, identifying 'Eight Americas' with gaps exceeding 20 years. Chetty et al. (2016) documented large differences in life expectancy across US counties, with race and income as key correlates. Kaufman et al. (2019) showed that Black-White differences in life expectancy varied substantially across four US states from 1969 to 2013. Roesch et al. (2023) found that the Black-White gap differed markedly across three large US cities, with male gaps on the order of 9 to 11 years. These studies demonstrate that state or national averages are inadequate for understanding local health disparities.

A less-explored dimension of this heterogeneity is the role of population composition. When a state's minority population is geographically concentrated in areas with poor health outcomes, the statewide disparity metric conflates two distinct phenomena: the underlying severity of local health disadvantage and the share of the population exposed to it. A counterfactual framework helps disentangle these. By hypothetically replacing one region's age-specific mortality rates with those of another while holding population shares constant, one can estimate how much of the statewide gap is attributable to conditions in a specific locality.

This compositional lens is especially relevant for Wisconsin. With about two-thirds of its Black population in Milwaukee County (a pattern shared by many states; Appendix C), any statewide Black health statistic is disproportionately a Milwaukee statistic. Understanding this is essential for correctly targeting interventions: statewide programs may dilute resources across regions where the gap is modest, while the community most in need -- Milwaukee -- requires concentrated investment.

Similarly, the interpretation of large Black-White gaps in high-amenity communities like Dane County (Madison) requires care. A large gap can arise from severely disadvantaged Black mortality, from exceptionally advantaged White mortality, or from both. These scenarios have different policy implications: the first calls for investment in Black health infrastructure; the second may simply reflect a comparison group with unusually favorable conditions.

3. Data and Methods

3.1 Data Sources

Mortality data come from individual death records for all deaths in Wisconsin from 2005 to 2017, including year of death, county of residence (FIPS code), age, sex, race, and Hispanic origin. I restrict the analysis to non-Hispanic Black and non-Hispanic White individuals (classified as non-Hispanic on the death record). The data contain 614,954 deaths meeting these criteria.

Population denominators come from the CDC's bridged-race population estimates for Wisconsin counties, 2005--2017, by single year of age (0--89, 90+), sex, and race. Hispanic origin is not separately identified in the population data (origin = 9 for all records), so the denominators include Hispanic individuals within each racial category. This is a minor concern for Black populations in Wisconsin, where the Hispanic share is small.

Geographic regions are defined using county FIPS codes: Milwaukee County (55079), Dane County (55025), and the rest of Wisconsin (all other counties).

Data management and analysis steps followed guidelines outlined in Sant'Anna (2024).

3.2 Life Table Construction

I construct abridged period life tables following Chiang (1968) and Preston et al. (2001). Deaths and populations are aggregated into 19 age groups: <1, 1--4, 5--9, ..., 80--84, 85+. To stabilize rates for smaller populations, I pool data over three-year windows centered on each year from 2006 to 2016 (e.g., the "2010" estimate uses data from 2009--2011). Both deaths and population are summed across the three-year window, so the resulting rates represent average annual age-specific mortality. These moving windows overlap substantially, so adjacent estimates should be interpreted as smoothed trends over

time rather than as independent year-by-year values. Infant nax values use Coale-Demeny formulas from Preston et al. (2001, Table 3.3); the open-ended interval (85+) uses nax = 1/M.

3.3 Counterfactual Framework

To decompose the sources of geographic variation, I conduct three counterfactual exercises. In each, I replace one group's set of 19 age-specific mortality rates with those of a donor region, hold the target population's age composition and size constant, and reconstruct the relevant life table. These are purely mechanical exercises -- they quantify the arithmetic importance of particular mortality schedules, not causal effects.

Exercise 1: I replace Milwaukee's Black age-specific mortality rates with those of the rest of Wisconsin's Black population, then recompute the statewide Black life table. This answers: how much of the statewide gap is driven by Milwaukee's elevated Black mortality? Exercise 2 substitutes Dane County's Black mortality rates for Milwaukee's, as an alternative benchmark. Exercise 3 asks a different question about Dane County: I replace Dane County's White mortality rates with those of the rest of the state, to estimate how much of Dane's gap reflects its exceptionally healthy White population rather than its Black mortality disadvantage.

3.4 Formal Gap Decomposition

To provide a comprehensive accounting of the statewide gap, I also decompose it into geography-specific contributions. If life expectancy is approximately a population-weighted average of sub-regional values, then the statewide Black-White gap can be written as:

Gap_WI ~= Sum_g [pi(g) x gap(g)] + W_comp + R

where pi(g) is the share of the statewide Black population living in geography g, gap(g) is the local Black-White life expectancy gap in geography g, W_comp is an adjustment because statewide White life expectancy is weighted by White (rather than Black) population shares, and R is a small residual term that reflects the nonlinearity of life expectancy aggregation. Each geography's contribution pi(g) x gap(g) is the product of a composition weight (how many Black residents live there) and a place-specific gap (how large the local disparity is). This decomposition quantifies how much of the statewide gap is attributable to each region and separates "place" effects from "composition" effects.

To isolate the composition effect, I compare the actual weighted sum of local gaps (using Black population shares) with the sum that would obtain if Black people had the same geographic distribution as White people. The difference measures how much the concentration of Black residents in high-gap regions inflates the statewide disparity beyond what geography-specific health conditions alone would produce.

3.5 Bootstrap Confidence Intervals

To quantify uncertainty in life expectancy and gap estimates, I use a parametric bootstrap. For each of 500 replicates, I resample death counts in each age-geography-race-sex cell from a Poisson distribution with mean equal to the observed death count, hold population denominators fixed, and recompute all life tables. Reported 95% confidence intervals are the 2.5th and 97.5th percentiles of the bootstrap distribution. This approach reflects sampling uncertainty in death counts; estimates for small populations (e.g., Dane County Black residents) are less precise. Confidence intervals are reported for life

expectancy levels, Black-White gaps, and changes in the gap over time. Decomposition contributions and counterfactual results are reported as point estimates without confidence intervals, because the bootstrap would need to jointly resample all geographic cells and propagate through the full decomposition machinery; this extension is left for future work.

3.6 Sensitivity and Uncertainty

The estimates presented below should be interpreted with sampling uncertainty and population size in mind. Bootstrap confidence intervals primarily capture variability in death counts given fixed population denominators; they do not account for potential systematic error in the numerator or denominator data. Geographic units with small Black populations, especially Dane County, have wider intervals and occasional apparent reversals over time that are consistent with noise. The figures and tables that report 95 percent intervals are intended to distinguish patterns that are well supported by the data from those that are more tentative. In particular, claims about worsening Black mortality in Dane County should be read in light of the reported uncertainty intervals.

4. Results

4.1 Geographic Variation in the Black-White Gap

Table 1 presents life expectancy and the Black-White gap for the most recent period (2015--2017) across the four geographic units. The results reveal substantial heterogeneity masked by the statewide average. (Statewide trends are validated against Roberts et al. 2019 in Appendix B.)

Table 1. Life expectancy at birth (years) by race, sex, and geography, 2015--2017 (3-year pooled).

Geography	Sex	White LE	Black LE	B-W Gap
Dane	Female	83.73	78.30	5.43
Dane	Male	80.21	71.71	8.50
Milwaukee	Female	82.05	75.95	6.10
Milwaukee	Male	77.48	68.25	9.23
Rest of WI	Female	82.13	77.81	4.33
Rest of WI	Male	77.78	73.94	3.84
Statewide	Female	82.25	76.53	5.72
Statewide	Male	77.95	70.03	7.91

For males, the Black-White gap ranges from 3.8 (95% CI 2.6--4.9) years in the rest of Wisconsin to 9.2 (95% CI 8.6--9.9) years in Milwaukee County -- a nearly 2.5-fold difference. Dane County's male gap (8.5 (95% CI 1.4--10.6) years) is nearly as large as Milwaukee's, despite Dane County's substantially higher overall life expectancy. This reflects a dual pattern: White males in Dane County enjoy the highest life expectancy of any group (80.2 years), but Black males in Dane County do not share proportionally in this advantage (71.7 years). Milwaukee has the lowest Black male life expectancy of any region (68.3 years), which is 5.7 years below the rest-of-state average for Black males.

For females, the geographic pattern is less dramatic but still notable. The gap ranges from 4.3 (95% CI 3.0--5.5) years (rest of Wisconsin) to 6.1 (95% CI 5.4--6.7) years (Milwaukee). Dane County's female

gap (5.4 (95% CI 3.3--7.3) years) is intermediate, but it features relatively high Black female life expectancy (78.3 years, the highest of any region). A caveat is warranted: Dane County's Black population is small relative to the other regions (roughly 50,000 per sex over the 3-year window, or about 103,000 total person-years, with approximately 260 male and 158 female Black deaths), which limits the precision of these estimates.

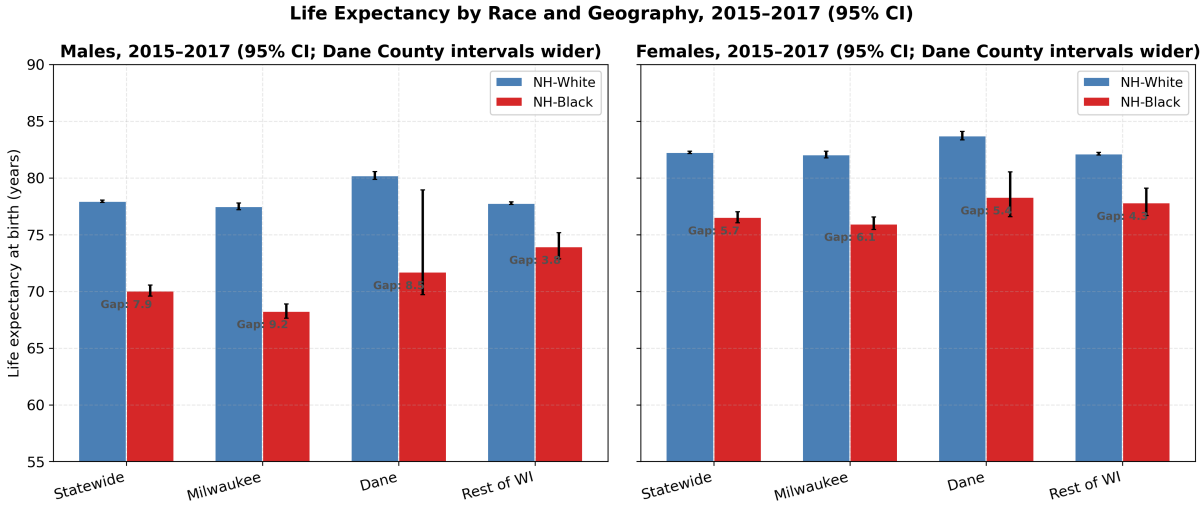

Figure 1. Life expectancy at birth by race and geography, males (left) and females (right), 2015--2017 (95% CI; Dane County intervals wider).

4.2 Trends in the Gap by Geography

Figures 2 and 3 plot the Black-White gap over time for each geography. The most striking finding is the divergent trajectory of the male gap across regions. In Milwaukee, the male gap narrowed from 8.6 years (2006) to a low of 7.1 years (2010) before widening sharply to 9.2 years by 2016. In Dane County, the male gap nearly doubled from 5.3 years (2006) to 8.5 years (2016). In the rest of Wisconsin, the gap remained comparatively small but also widened, from 2.1 to 3.8 years.

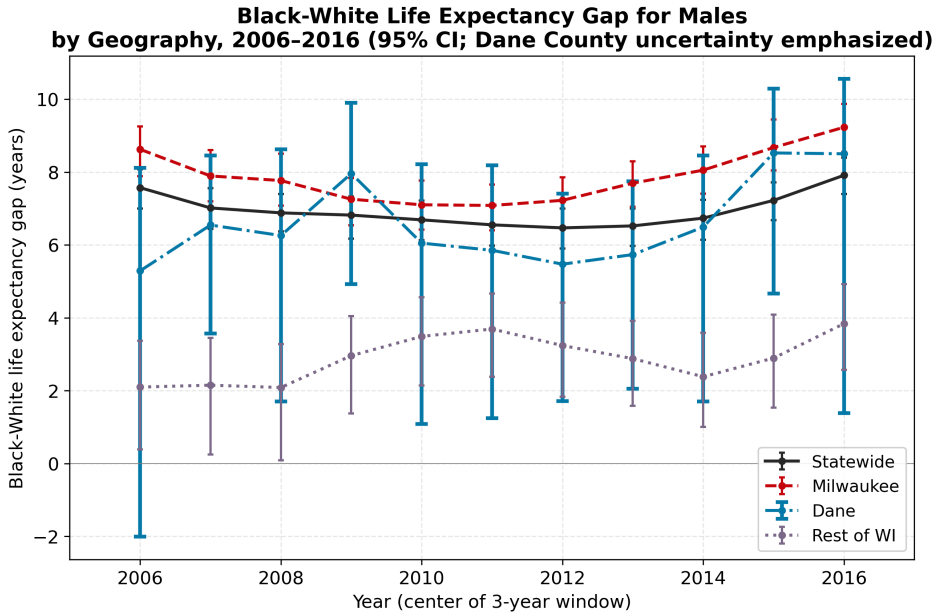

Figure 2. Black-White life expectancy gap for males by geography, 2006--2016 (3-year pooled; 95% CI; Dane County uncertainty emphasized).

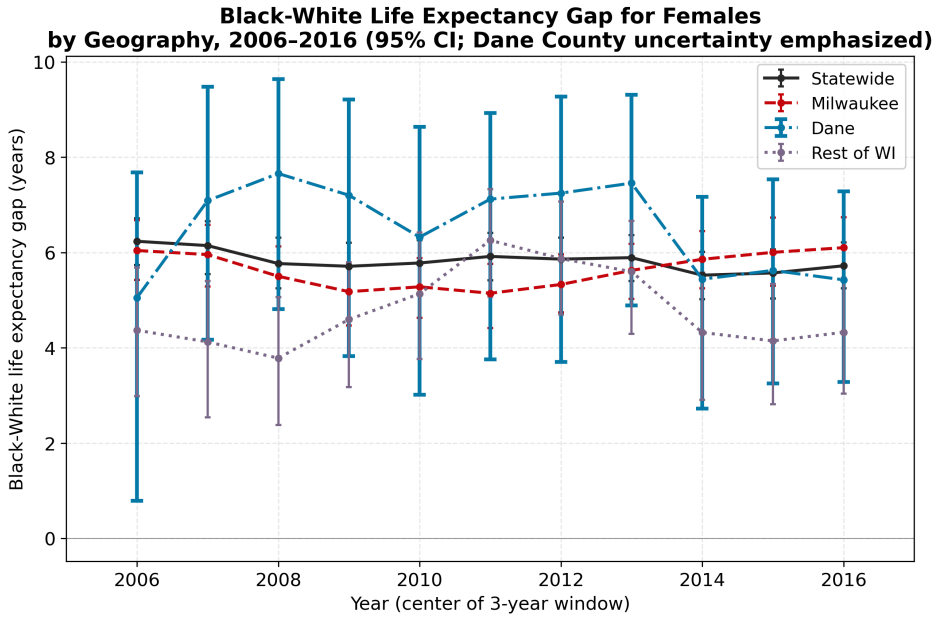

Figure 3. Black-White life expectancy gap for females by geography, 2006--2016 (3-year pooled; 95% CI; Dane County uncertainty emphasized).

For females, trends were more stable. The statewide female gap narrowed slightly from 6.2 to 5.7 years. Milwaukee's female gap remained roughly constant near 6 years. Dane County showed considerable volatility -- likely reflecting the small Black female population -- but ended the period at 5.4 years, similar to where it began.

4.3 Decomposing the Change Over Time

Table 2 decomposes the change in the gap between the earliest (2005--2007) and latest (2015--2017) periods into changes in White and Black life expectancy. A widening gap could reflect White gains, Black losses, or both -- and the pattern differs sharply by geography.

Table 2. Change in the Black-White life expectancy gap (years) between 2005--2007 and 2015--2017, with decomposition into White and Black LE changes. 95% CI in brackets when available.

Geography	Sex	Gap (05-07)	Gap (15-17)	Change (95% CI)	Delta White LE	Delta Black LE (95% CI)
Statewide	Male	7.57	7.91	0.34 [-0.4, 1.1]	0.55	0.20 [-0.6, 0.9]
Statewide	Female	6.24	5.72	-0.52 [-1.2, 0.2]	0.20	0.72 [-0.0, 1.4]
Milwaukee	Male	8.62	9.23	0.61 [-0.4, 1.6]	1.13	0.52 [-0.3, 1.4]
Milwaukee	Female	6.04	6.10	0.06 [-0.8, 0.9]	0.77	0.71 [-0.1, 1.5]
Dane	Male	5.29	8.50	3.21 [-4.1, 11.3]	0.85	-2.37 [-10.4, 5.0]
Dane	Female	5.05	5.43	0.38 [-3.3, 5.2]	0.46	0.08 [-4.7, 3.8]
Rest of WI	Male	2.10	3.84	1.74 [-0.2, 3.9]	0.41	-1.33 [-3.4, 0.5]
Rest of WI	Female	4.37	4.33	-0.04 [-1.8, 1.8]	0.07	0.12 [-1.8, 1.9]

In Dane County, the male gap widened by 3.2 years -- the largest increase of any region-sex combination. The estimates point to a decline of about 2.4 years in Black male life expectancy (from 74.1 to 71.7 years), while White male life expectancy increased by 0.9 years. The estimated decline in Dane County Black male life expectancy is -2.4 years (95% CI -10.5 to 5.0). The interval includes zero; precision is limited by the small Black male population in Dane County. In Milwaukee, the male gap widened by 0.6 years, with both groups gaining but White males gaining faster (+1.1 vs. +0.5 years). In the rest of Wisconsin, Black male life expectancy declined by 1.3 years, widening the gap by 1.7 years. Among females, the statewide gap narrowed by 0.5 years, driven by faster improvement among Black females (+0.7 years) relative to White females (+0.2 years).

4.4 Decomposing the Statewide Gap

The formal decomposition reveals that the statewide gap is overwhelmingly driven by Milwaukee. Table 3 presents the decomposition for three key periods and both sexes. Each geography's contribution equals the product of its Black population share (pi) and its local B-W gap. The contributions sum, with a small White composition adjustment, to the total statewide gap.

Table 3. Decomposition of the statewide B-W gap (years). Each geography's contribution = pi_g (Black pop. share) x local B-W gap. Percentages show share of total gap. Adj. = White composition + aggregation residual.

Year	Sex	Gap	Milwaukee	Dane	Rest of WI	Adj.
2006	Male	7.57	5.67 (75%)	0.38 (5%)	0.57 (7%)	0.95
2006	Female	6.24	4.45 (71%)	0.33 (5%)	0.87 (14%)	0.59
2010	Male	6.69	4.53 (68%)	0.47 (7%)	0.99 (15%)	0.69
2010	Female	5.78	3.78 (65%)	0.46 (8%)	1.08 (19%)	0.46
2016	Male	7.91	5.58 (71%)	0.72 (9%)	1.19 (15%)	0.42
2016	Female	5.72	4.15 (72%)	0.43 (7%)	1.05 (18%)	0.10

For males in 2016, Milwaukee contributes 5.6 years of the 7.9 (95% CI 7.4--8.4)-year statewide gap -- 70 percent of the total. This dominance reflects both Milwaukee's large Black male population share (about 60 percent) and its large local gap (9.2 (95% CI 8.6--9.9) years). Dane County's gap is nearly as large (8.5 (95% CI 1.4--10.6) years), but because only 8.5 percent of the state's Black males live there, its contribution is just 0.7 years (9 percent). The rest of Wisconsin contributes 1.2 years (15 percent), with a moderate gap (3.8 years) offset by a moderate population share (31 percent). The White composition adjustment is small (0.4 years combined), confirming that the decomposition is driven almost entirely by the interplay of Black population shares and local gaps.

For females, Milwaukee's contribution is also dominant (4.1 years, 73 percent of the 5.7-year gap in 2016), though the rest of Wisconsin contributes a larger share (18 percent) than for males, reflecting the fact that the female gap is more geographically uniform.

Figure 4 visualizes this decomposition over time. Each stacked bar shows how the statewide gap is built from geography-specific contributions. Milwaukee's red segment dominates in every period, but its share of the total has fluctuated. For males, the widening gap after 2012 reflects growing contributions from all three regions, with both Milwaukee's and the rest of Wisconsin's segments expanding.

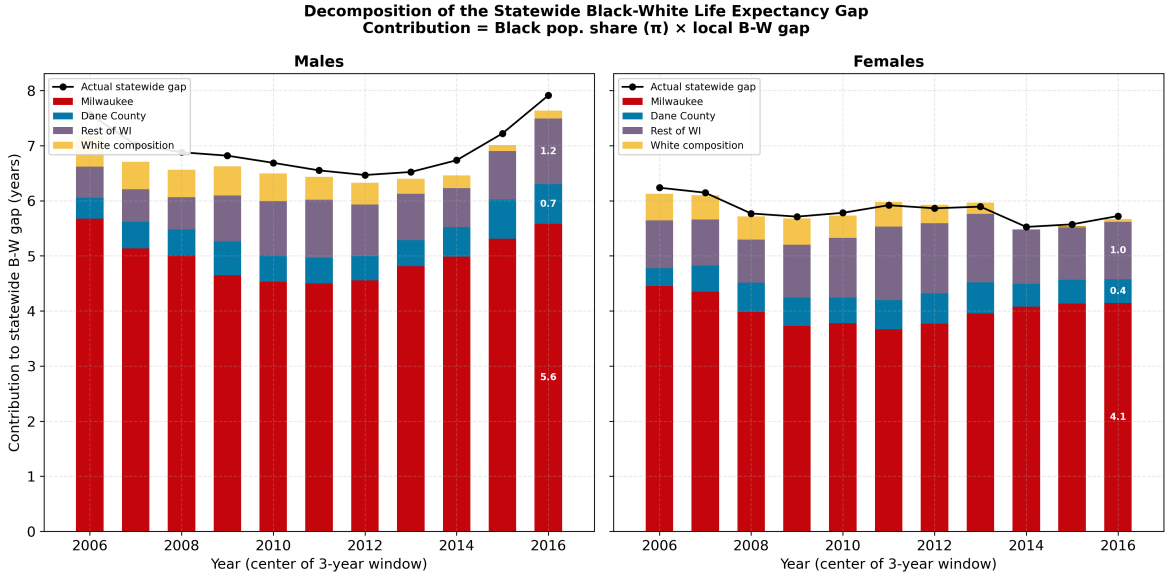

Figure 4. Decomposition of the statewide Black-White life expectancy gap into geographic contributions (pi x local gap) for males (left) and females (right), 2006--2016.

The composition effect -- how much the geographic concentration of Black residents in high-gap areas inflates the statewide gap -- is substantial. If Black Wisconsinites had the same geographic distribution as White Wisconsinites, the weighted sum of local male gaps would be 4.9 years instead of 7.5 years in 2016: the composition penalty is 2.6 years. In other words, roughly a third of what drives the statewide male gap is not the severity of local disparities per se, but the fact that most Black residents live in the worst-performing region.

Figure 5 illustrates the two components of each geography's contribution: the Black population share (Panel A) and the local B-W gap (Panel B). Milwaukee's contribution is large because it ranks highest on both dimensions. Dane County's high local gap is offset by its small Black population share, keeping its statewide contribution modest.

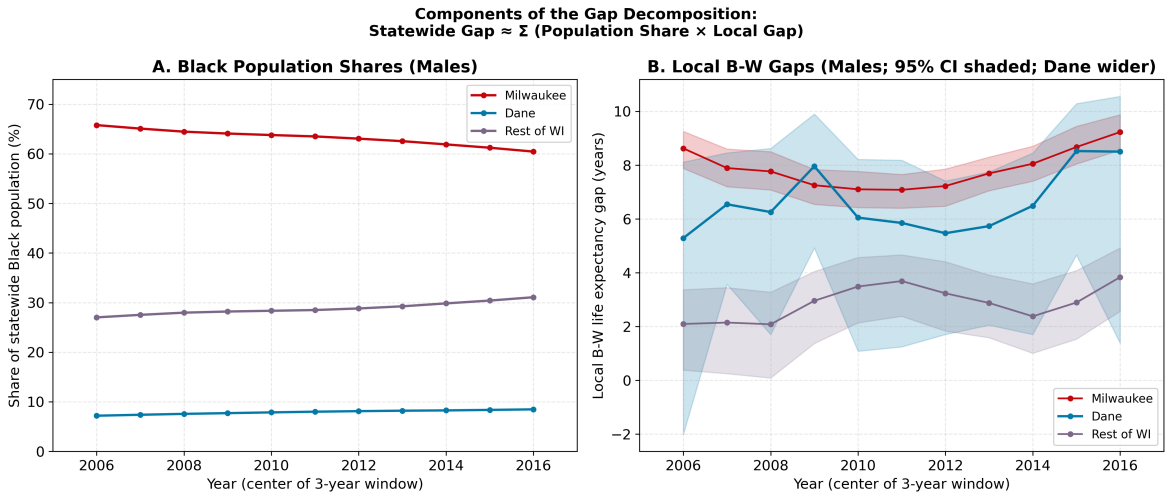

Figure 5. Components of the gap decomposition for males: (A) Black population shares by geography and (B) local B-W life expectancy gaps by geography, 2006--2016.

4.5 Counterfactual Exercises

The decomposition quantifies each region's contribution to the statewide gap; the counterfactual exercises below ask what the gap would be under specific alternative mortality schedules.

Table 4 presents Exercise 1: what if Black Milwaukeeans had the same age-specific mortality rates as Black people in the rest of Wisconsin? In 2016, the statewide male gap would fall from 7.9 (95% CI 7.4--8.4) to 4.2 years -- a reduction of 3.75 years, or nearly half the observed gap. Black male life expectancy would rise from 70.0 to 73.8 years. For females, the gap would narrow from 5.7 to 4.4 years. Note that this ~47 percent counterfactual reduction differs from the 70 percent decomposition share reported in Table 3: the former measures the fractional reduction in the gap under a specific mortality-replacement scenario, while the latter measures Milwaukee's share of the gap in a population-weighted accounting. Both point to Milwaukee as the dominant driver, but they quantify different aspects of its role.

Table 4. Counterfactual Exercise 1: statewide B-W gap if Milwaukee's Black population had Rest-of-WI Black age-specific mortality.

Year	Sex	Actual Gap	CF Gap	Reduction	Actual Black e0	CF Black e0

2006	Male	7.57	2.19	5.38	69.83	75.21
2006	Female	6.24	4.32	1.91	75.81	77.73
2010	Male	6.69	3.64	3.05	71.38	74.43
2010	Female	5.78	5.16	0.62	76.87	77.49
2016	Male	7.91	4.17	3.75	70.03	73.78
2016	Female	5.72	4.37	1.35	76.53	77.88

Exercise 2 asks: what if Milwaukee's Black population instead had Dane County's Black mortality? In 2006, this would have reduced the male gap by 4.6 years (from 7.6 to 3.0), since Dane County Black males had relatively favorable life expectancy at that time (74.1 years). But by 2016, this exercise yields a smaller reduction of only 2.3 years (from 7.9 to 5.6), because Dane County Black male mortality had itself deteriorated. This time comparison underscores that the Black male mortality crisis extends beyond Milwaukee.

Exercise 3 addresses a different question: why is Dane County's gap so large despite its reputation as a progressive, high-opportunity community? Part of the answer is that the comparison group -- Dane County Whites -- is exceptionally healthy. Table 5 shows what happens when Dane County White mortality is replaced with Rest-of-WI White mortality.

Table 5. Counterfactual Exercise 3: Dane County B-W gap if Dane Whites had Rest-of-WI White age-specific mortality.

Year	Sex	Actual Gap	CF Gap	Reduction	Actual White e0	CF White e0
2006	Male	5.29	3.29	2.00	79.36	77.37
2006	Female	5.05	3.84	1.21	83.27	82.06
2010	Male	6.05	4.32	1.73	79.76	78.04
2010	Female	6.32	5.03	1.30	83.91	82.61
2016	Male	8.50	6.07	2.43	80.21	77.78
2016	Female	5.43	3.83	1.59	83.73	82.13

In 2016, replacing Dane White mortality with Rest-of-WI rates reduces the male gap from 8.5 to 6.1 years and the female gap from 5.4 to 3.8 years. Roughly one-quarter to one-third of Dane County's gap is attributable to the exceptionally favorable White mortality environment. The remaining gap of 6.1 years for males is still substantial and suggests disadvantaged Black mortality in Dane County. Point estimates in Table 2 suggest a 2.4-year decline in Black male life expectancy, although the wide confidence interval means this should be interpreted cautiously. Dane County's large gap thus reflects an unusually healthy White benchmark combined with suggestive evidence of worsening Black health outcomes.

5. Discussion

This analysis yields three principal findings. First, the statewide Black-White life expectancy gap in Wisconsin masks a nearly 2.5-fold range across regions: from 3.8 (95% CI 2.6--4.9) years in the rest of the state to 9.2 (95% CI 8.6--9.9) years in Milwaukee for males. Second, the formal decomposition reveals that Milwaukee contributes 70 percent of the statewide male gap -- a dominance driven by the combination of its large Black male population share (about 60 percent in 2016) and its large local gap (9.2 (95% CI 8.6--9.9) years). The geographic concentration of Black Wisconsinites in Milwaukee adds

approximately 2.6 years to the statewide male gap compared to a scenario in which Black and White residents had the same geographic distribution. Third, the male gap widened in all regions between 2005-2007 and 2015-2017. A cautionary note concerns Dane County, where the male gap (8.5 (95% CI 1.4--10.6) years) is nearly as large as Milwaukee's. Counterfactual analysis shows that roughly a quarter to a third of Dane's gap reflects its exceptionally healthy White population rather than Black disadvantage per se. Point estimates also suggest deteriorating Black male mortality in Dane County, but this pattern is estimated with substantial uncertainty due to the small Black population and should be interpreted as a signal warranting further investigation rather than an established trend.

These findings have important implications for how we interpret and respond to state-level racial health disparities. The "Wisconsin gap" is not a diffuse, statewide phenomenon amenable to generic state-level programs. The decomposition shows that 70 percent of the statewide male gap (7.9 (95% CI 7.4--8.4) years) traces to a single county where about two-thirds of the state's Black population lives under mortality conditions that produce a male life expectancy of 68.3 years -- below the US national average for Black males and 5.7 years below the Black male average in the rest of the state. Statewide interventions that spread resources uniformly are poorly matched to this geographic concentration of disadvantage.

The Dane County finding deserves particular attention. Dane County is widely perceived as a progressive, high-opportunity community with strong public institutions and a robust economy. Yet the Black-White male life expectancy gap there (8.5 (95% CI 1.4--10.6) years) is nearly as large as Milwaukee's, and it widened faster. The counterfactual decomposition reveals that this gap has two sources: the exceptional longevity of Dane County's White population (which inflates the reference standard by 2.4 years) and what point estimates suggest is a decline in Black male life expectancy, though this is imprecisely estimated given the small population. The latter is the more actionable concern. Favorable average conditions do not automatically produce equitable health outcomes. Social determinants driving mortality disparities may operate through mechanisms -- residential segregation within the county, differential incarceration, concentrated poverty -- that are not captured by county-level aggregates.

The decline in Black male life expectancy in several regions during the latter years of the study period is consistent with emerging national trends linked to the opioid epidemic, increases in homicide, and rising "deaths of despair" (Case & Deaton 2015). Cause-of-death decomposition (e.g., Arriaga 1984) is needed to identify which specific causes are driving these reversals and to inform targeted interventions.

Several limitations should be noted. The population denominators do not separately identify Hispanic origin, which may introduce a small upward bias in population counts for non-Hispanic groups. Reported 95% confidence intervals from a parametric bootstrap (Section 3.5) reflect sampling uncertainty in death counts; Dane County's Black population, while growing, remains relatively small, and the 3-year pooling mitigates but does not eliminate instability in those estimates. As a result, the apparent decline in Black male life expectancy in Dane County should be viewed as a warning signal that warrants continued monitoring and investigation rather than as a precisely measured effect. The counterfactual exercises are mechanical decompositions, not causal estimates -- they quantify the arithmetic contribution of regional mortality schedules but do not identify why those schedules differ. The analysis covers 2005--2017 and does not capture more recent trends, including the COVID-19 pandemic, which disproportionately affected Black life expectancy nationwide. Finally, this analysis does not decompose the gap by cause of death, which would help identify targeted intervention points.

6. Conclusion

Statewide life expectancy statistics, while valuable, can obscure the geographic concentration of racial health disparities. In Wisconsin, the Black-White life expectancy gap (7.9 (95% CI 7.4--8.4) years for males) varies by a factor of nearly 2.5 across regions, and the trajectories are diverging. A formal decomposition reveals that Milwaukee alone accounts for 70 percent of the statewide male gap, driven by the double burden of a large Black population share and severe local mortality disadvantage. Black geographic concentration adds roughly 2.6 years to the statewide male gap -- a quantifiable composition penalty. Dane County's gap (8.5 (95% CI 1.4--10.6) years), meanwhile, is inflated by exceptionally high White life expectancy but also reflects what appears to be a troubling decline in Black male life expectancy, a finding that warrants continued monitoring given the small population size. Policy responses must be geographically targeted: Milwaukee's crisis requires investment in the structural determinants of Black health in a hyper-segregated urban environment, while Dane County's growing disparity calls for investigation of why a seemingly prosperous community is failing its Black residents.

Appendix A: Geographic Reference

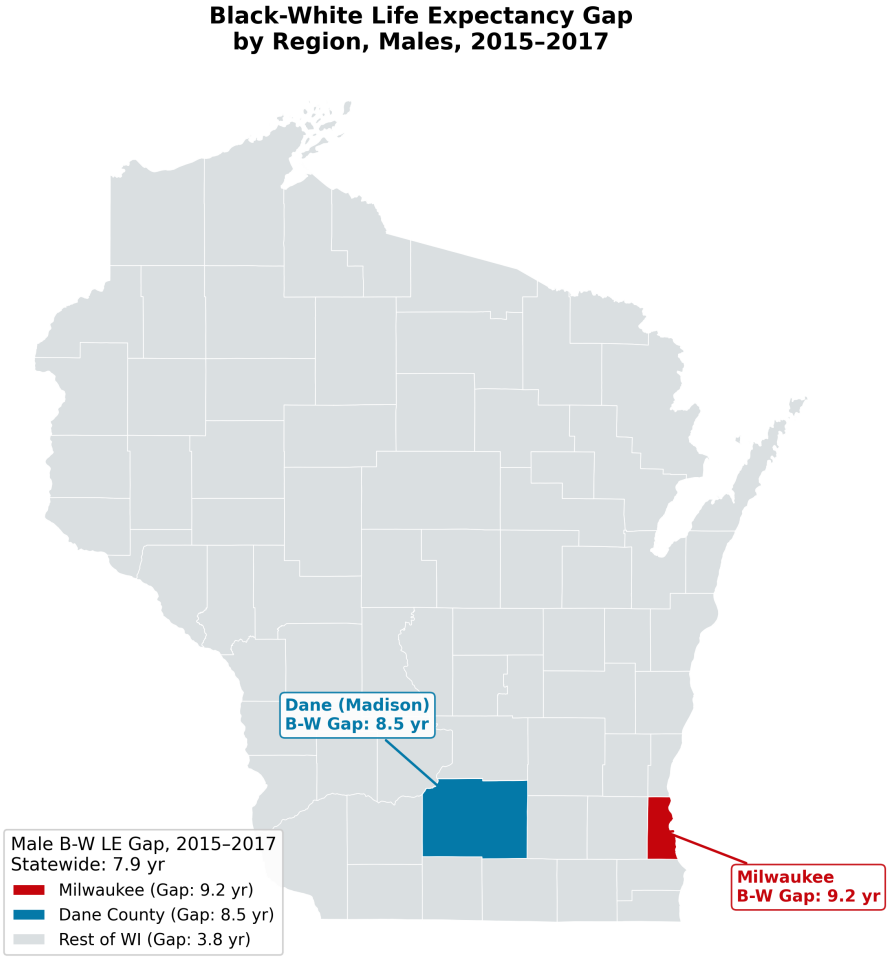

Figure A1. Wisconsin counties with Milwaukee, Dane, and rest-of-state regions highlighted. Annotations show male Black-White life expectancy gap for 2015--2017.

Appendix B: Replication of Roberts et al. (2019)

To validate the analytic pipeline, I replicate the statewide life expectancy estimates of Roberts et al. (2019) for the 2014--2016 period (center year 2015). Table B1 compares my estimates with the published values. My estimates are within 0.1 to 0.4 years of the published values across all race-sex groups. The small discrepancies likely reflect differences in Hispanic-origin filtering and the exact population denominators used. The Black-White gap in my estimates is 7.2 years for males and 5.6 years for females, compared to 7.3 and 5.6 years in Roberts et al.

Table B1. Replication check: statewide life expectancy (years) for 2014--2016, compared with Roberts et al. (2019).

Group	Roberts et al.	This Paper	Difference
White Males	77.75	78.06	+0.31
Black Males	70.41	70.84	+0.43
White Females	82.15	82.24	+0.09
Black Females	76.54	76.67	+0.13
Male B-W Gap	7.34	7.22	-0.12
Female B-W Gap	5.61	5.57	-0.04

Figures B1 and B2 show statewide life expectancy trends for males and females, respectively. Among males, White life expectancy was essentially flat (77.4 to 78.0 years), while Black male life expectancy rose from 69.8 to a peak of 71.6 years around 2012 before declining to 70.0 years by 2016. This reversal widened the male gap from 6.5 years at its narrowest (2012) to 7.9 years at the end of the study period. Female trends were more stable, with the gap fluctuating between 5.6 and 6.2 years.

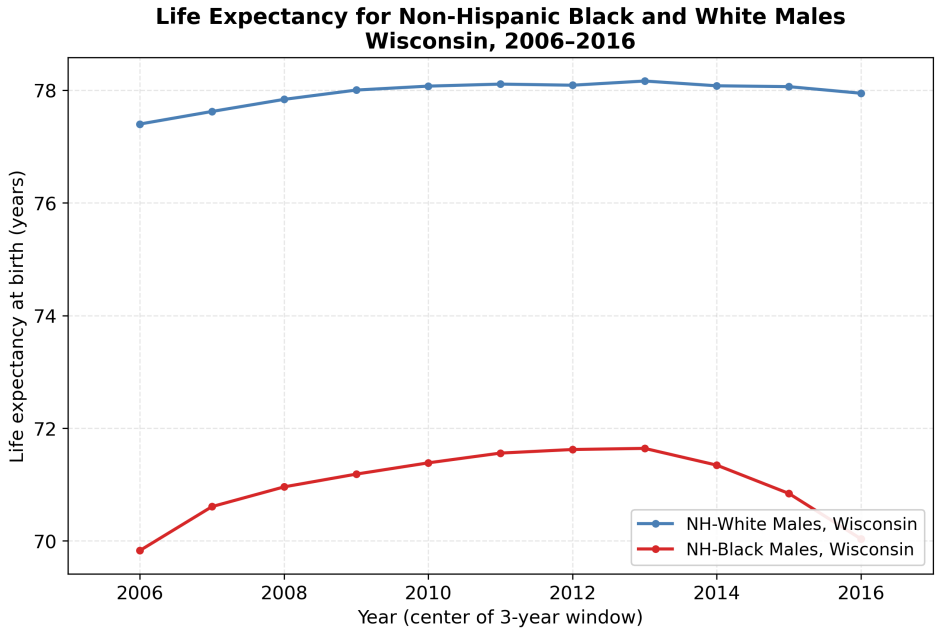

Figure B1. Life expectancy at birth for non-Hispanic Black and White males, Wisconsin statewide, 2006--2016 (3-year pooled estimates).

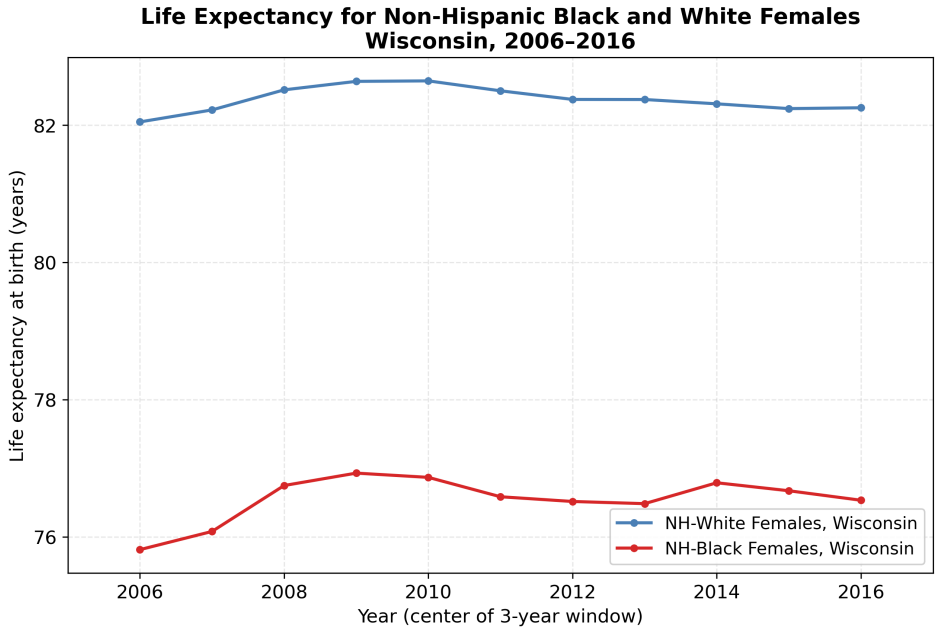

Figure B2. Life expectancy at birth for non-Hispanic Black and White females, Wisconsin statewide, 2006--2016 (3-year pooled estimates).

Appendix C: Sample Context and County Concentration

C.1 Sample context by geography, sex, and race

Table C1 reports the number of deaths, pooled 3-year person-years, and crude death rates by geography, sex, and race for 2015--2017 (the 2016 centered window). These figures provide context for the precision of life expectancy estimates: cells with fewer deaths and smaller person-years, especially Black residents in Dane County, yield wider confidence intervals and more sampling variability.

Table C1. Sample context for 2015--2017 (center year 2016): deaths, pooled person-years, and crude death rates by geography, sex, and race.

Geography	Sex	Race	Deaths (3yr)	Person-years (3yr)	Crude rate (per 1,000)
Dane	Female	Black	158	50447	3.132
Dane	Female	White	4697	699209	6.718
Dane	Male	Black	260	52777	4.926
Dane	Male	White	4378	693702	6.311
Milwaukee	Female	Black	2838	434594	6.530
Milwaukee	Female	White	8993	946700	9.499
Milwaukee	Male	Black	3243	376514	8.613
Milwaukee	Male	White	8244	923921	8.923
Rest of WI	Female	Black	670	154678	4.332
Rest of WI	Female	White	57623	6061187	9.507
Rest of WI	Male	Black	917	193571	4.737
Rest of WI	Male	White	58861	6087385	9.669
Statewide	Female	Black	3666	639719	5.731
Statewide	Female	White	71313	7707096	9.253
Statewide	Male	Black	4420	622862	7.096
Statewide	Male	White	71483	7705008	9.277

C.2 County concentration of Black population by state

Wisconsin's concentration of its Black population in Milwaukee County is not unusual. Many states have a single county or equivalent that holds the largest share of the state's Black population. Table C2 shows, for each state and the District of Columbia, the county (or county-equivalent) with the largest share of the state's Black population and that share. Wisconsin ranks among the states where this concentration is high: Milwaukee County holds about two-thirds of the state's Black population. Several states show even higher concentration (e.g., Nevada 94%, Hawaii 89%, Rhode Island 87%, Arizona 77%, Illinois 67%). Thus the compositional challenge for interpreting statewide disparity statistics is widespread, not unique to Wisconsin.

Table C2. County (or county-equivalent) with largest share of state's Black population and approximate share. Source: compiled from publicly available demographic estimates following Sant'Anna (2024) workflow guidance.

State	County (largest share)	Share (%)

Alabama	Jefferson County Alabama	21.7%
Alaska	Anchorage Municipality Alaska	66.1%
Arizona	Maricopa County Arizona	77.3%
Arkansas	Pulaski County Arkansas	32.6%
California	Los Angeles County California	35.5%
Colorado	Arapahoe County Colorado	30.7%
Connecticut	Capitol Planning Region Connecticut	32.1%
Delaware	New Castle County Delaware	66.2%
District of Columbia	District of Columbia	100.0%
Florida	Broward County Florida	16.4%
Georgia	Fulton County Georgia	13.7%
Hawaii	Honolulu County Hawaii	88.7%
Idaho	Ada County Idaho	49.0%
Illinois	Cook County Illinois	67.0%
Indiana	Marion County Indiana	42.8%
Iowa	Polk County Iowa	29.8%
Kansas	Sedgwick County Kansas	25.7%
Kentucky	Jefferson County Kentucky	47.6%
Louisiana	Orleans Parish Louisiana	14.8%
Maine	Cumberland County Maine	44.2%
Maryland	Prince George's County Maryland	31.8%
Massachusetts	Suffolk County Massachusetts	31.2%
Michigan	Wayne County Michigan	49.2%
Minnesota	Hennepin County Minnesota	44.2%
Mississippi	Hinds County Mississippi	15.0%
Missouri	St. Louis County Missouri	34.9%
Montana	Yellowstone County Montana	22.9%
Nebraska	Douglas County Nebraska	66.7%
Nevada	Clark County Nevada	93.7%
New Hampshire	Hillsborough County New Hampshire	52.2%
New Jersey	Essex County New Jersey	26.6%
New Mexico	Bernalillo County New Mexico	43.2%
New York	Kings County New York	26.7%
North Carolina	Mecklenburg County North Carolina	15.9%
North Dakota	Cass County North Dakota	47.7%
Ohio	Cuyahoga County Ohio	25.2%
Oklahoma	Oklahoma County Oklahoma	41.2%
Oregon	Multnomah County Oregon	55.2%
Pennsylvania	Philadelphia County Pennsylvania	45.4%
Rhode Island	Providence County Rhode Island	87.0%
South Carolina	Richland County South Carolina	14.8%

South Dakota	Minnehaha County South Dakota	63.3%
Tennessee	Shelby County Tennessee	44.2%
Texas	Harris County Texas	25.1%
Utah	Salt Lake County Utah	57.1%
Vermont	Chittenden County Vermont	53.3%
Virginia	Fairfax County Virginia	6.9%
Washington	King County Washington	48.8%
West Virginia	Kanawha County West Virginia	19.6%
Wisconsin	Milwaukee County Wisconsin	67.0%
Wyoming	Laramie County Wyoming	48.4%

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